

Problem C: Count Equation Solutions

Time Limit: 5 seconds

Description

Count the number of positive integer solutions to the following equation:

$$a_1 x_1 - a_2 x_2 + a_3 x_3 - a_4 x_4 + a_5 x_5 - a_6 x_6 = 0$$

where a_i are integer coefficients, and x_i are the variables such that $1 \leq x_i \leq M$.

Input

A number of inputs (≤ 600) described as follows. The first line is M on a single line, followed by the 6 coefficients a_i on the next line. Constraint is $1 \leq M \leq 100$ and $0 < a_i \leq 1000000$.

Output

For each input, output the number of solutions on a single line.

Sample Input

```
2
1 1 1 1 1 1
```

Sample Output

```
20
```